Task 1:

Task 1

```
#include <iostream>
using namespace std;
int main() {
char letter;
letter = 'A'
cout << letter<< "\n";
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
```

```
#include <iostream>
using namespace std;
int main() {
char letter
letter = 65;
cout << letter << '\n';
letter = 66;
cout << Letter << '\n';
cin >> letter;
return 0;
}
```

```
#include <iostream>
using namespace std;
int main () {
int letter;
letter = 'A';
cout << letter << '\n';
Letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
```

main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
int main(){
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Run ▶

Output

```
A
B
```

main.cpp

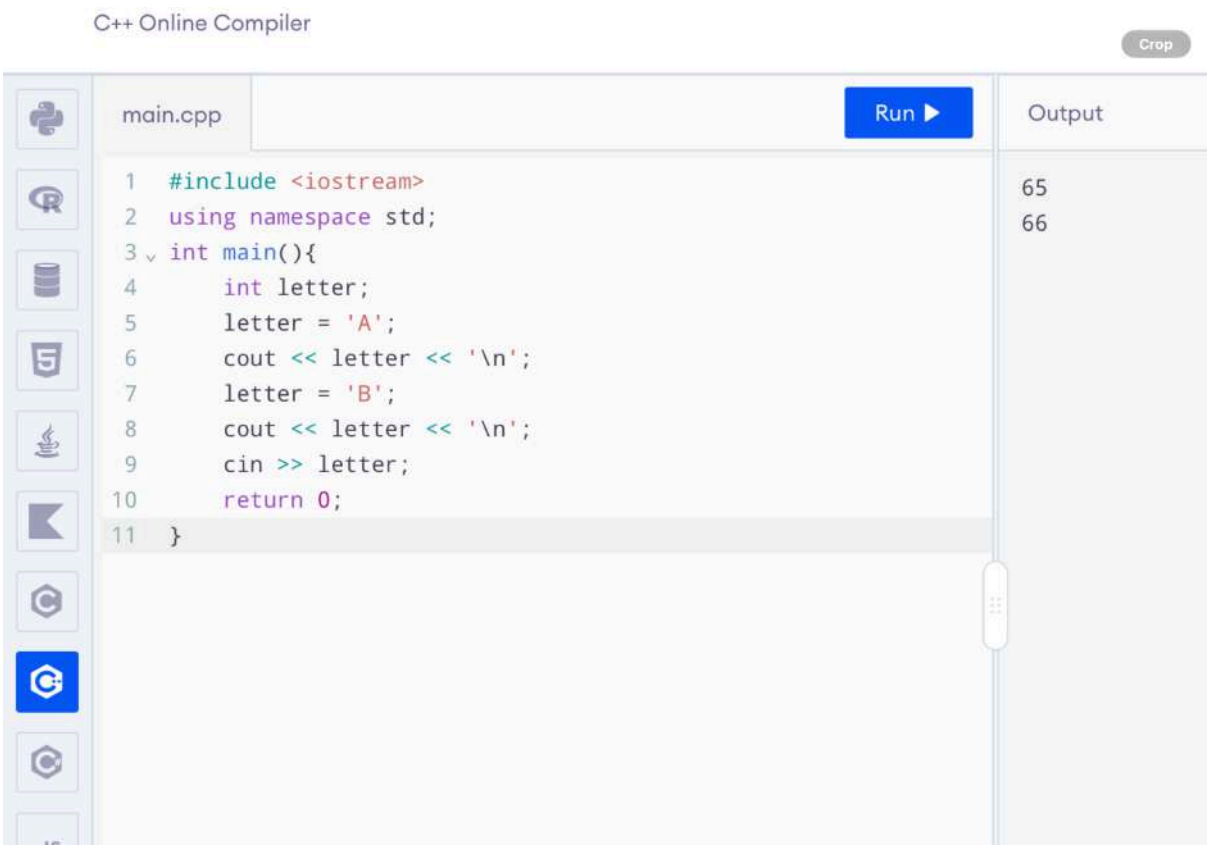
Run ▶

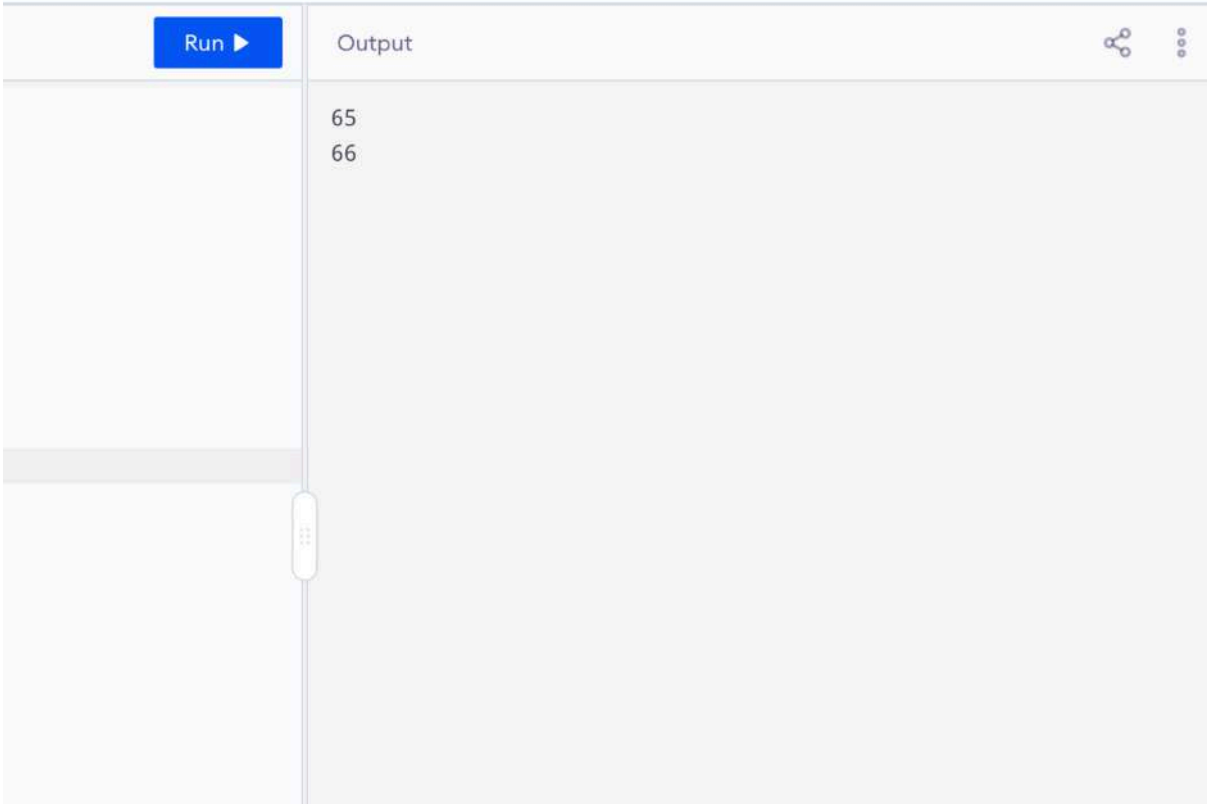
```cpp
#include <iostream>
using namespace std;
int main(){
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

A
B

Output

A
B


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main.cpp
Run
Output
#include <iostream>
using namespace std;
int main(){
int letter;
letter = 'A';
cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
65
66


Run
Output
65
66

Task 2:

Task 2

```
#include <iostream>
using namespace std;
int main () {
  int num1, num2;
cout << "Enter first number: ";
cin >> num1;
cout << "Inter second number: ";
cin >> num2;
cout << "Addition: " << num1 + num2 << endl;
cout << "Subtraction: " << num1 - num2 << endl;
cout << "Multiplication: " << num1 * num2 << endl;
cout << "Division: " << num1 / num2 << endl;
return 0;
}
```

main.cpp

Run ▶

Output

```cpp
#include <iostream>
using namespace std;
int main(){
    int num1, num2;
    cout << "Enter first number:";
    cin >> num1;
    cout << "Enter second number:";
    cin >> num2;
    cout << "Addition:" << num1 + num2 << endl;
    cout << "Subtraction:" << num1 - num2 << endl;
    cout << "Multiplication:" << num1 * num2 << endl;
    cout << "Division:" << num1 / num2 << endl;
    return 0;
}
```

Output

Crop

```
Enter first number:20
Enter second number:10
Addition:30
Subtraction:10
Multiplication:200
Division:2
```

Task 3:

Task 3

```
#include <iostream>
using namespace std;
int main() {
  int val1;
  int val2;
  int val3;
cout << "Please enter your 3 numbers : ";
cin >> val1 >> val2 >> val3;
cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
 system("PAUSE");
 return 0;
 }
```

Output

```
Please enter your 3 numbers : 10 20 30
10
20
30
30
20
10
```

main.cpp

Run

```cpp

#include <iostream>
using namespace std;
int main(){
    int val1;
    int val2;
    int val3;
    cout << "Please enter your 3 numbers:";
    cin >> val1 >> val2 >> val3;
    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
    system("PAUSE");
    return 0;
}
```

Output

```
Please enter your 3 numbers:10 20 30
10
20
30
30
20
10
```

Task 4:

Task 4

```
# include <iostream>
# include <cmath>
using namespace std;
int main(){
double x1, y1, x2, y2, distance;
cout << "Enter x1 and y1 : ";
cin >> x1 >> y1;
cout << "Enter x2 and y2 : ";
cin >> x2 >> y2;
distance = sqr(pow(x2-x1,2) + pow(y2-y1,2));
cout << "Distance between points : "<< distance << endl;
return 0;
}
```

main.cpp

Run ▶

```cpp

#include <iostream>
#include <cmath>
using namespace std;
int main(){
    double x1, y1, x2, y2, distance;
    cout << "Enter x1 and y1:";
    cin >> x1 >> y1;
    cout << "Enter x2 and y2:";
    cin >> x2 >> y2;
    distance = sqrt(pow(x2 - x1,2) + pow(y2 - y1,2));
    cout << "Distance between points:" << distance << endl;
    return 0;
}

```

Output

Crop

```
Enter x1 and y1:1 2
Enter x2 and y2:7 10
Distance between points:10
```

Task 5:

Task 5

```
#include <iostream>
using namespace std;
int main () {
const double PI = 3.14159;
double radius;
cout << "Enter radius of Circle : ";
cin >> radius;
cout << "Area of Circle : "<< PI * radius * radius << endl;
double length , width;
cout << "\n Enter length and width of rectangle : ";
cin >> length >> width ;
cout << "Area of Rectangle : "<< length * width << endl;
double base , height ;
cout << " \n Enter base and height of triangle : ";
cin >> base >> height ;
cout << "Area of Triangle : " << base * height * 0.5 << endl;
return 0;
}
```

main.cpp

Run ▶

Output

```cpp
#include <iostream>
using namespace std;
int main(){
    const double PI = 3.14159;
    double radius;
    cout << "Enter radius of circle:";
    cin >> radius;
    cout << "Area of circle: " << PI * radius * radius << endl;
    double length, width;
    cout << "\nEnter length and width of rectangle: ";
    cin >> length >> width;
    cout << "Area of Rextangle: " << length * width << endl;
    double base, height;
    cout << "\nEnter base and height of triangle: ";
    cin >> base >> height;
    cout << "Area of Triangle: " << base * height * 0.5 << endl;
    return 0;

}
```

Output

```
Enter radius of circle:2
Area of circle: 12.5664

Enter length and width of rectangle: 7 5
Area of Rextangle: 35

Enter base and height of triangle: 10 20
Area of Triangle: 100
```

Task 6:

Task 6

```
#include <iostream>
using namespace std;
int main () {
   double item1 = 12.95;
   double item2 = 24.95;
   double item3 = 6.95;
   double item4 = 14.95;
   double item5 = 3.95;
   double subtotal = item1 + item2 + item3 + item4 + item5;
   double salesTax = subtotal * 0.06;
   double total = subtotal + salesTax;
   cout << "Item 1 Price : $" << item1 << endl;
   cout << "Item 2 Price : $" << item2 << endl;
   cout << "Item 3 Price : $" << item3 << endl;
   cout << "Item 4 Price : $" << item4 << endl;
   cout << "Item 5 Price : $" << item5 << endl;
   cout << "subtotal :     $" << subtotal << endl;
   cout << "Sales Tax :    $" << salesTax << endl;
   cout << "Total :        $" << total << endl;
   return 0;
}
```


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main.cpp
Run
#include <iostream>
using namespace std;
int main() {
double item1 = 12.95;
double item2 = 24.95;
double item3 = 6.95;
double item4 = 14.95;
double item5 = 3.95;
double subtotal = item1 + item2 + item3 + item4 + item5;
double salesTax = subtotal * 0.06;
double total = subtotal + salesTax;
cout << "Item 1 Price: $" << item1 << endl;
cout << "Item 2 Price: $" << item2 << endl;
cout << "Item 3 Price: $" << item3 << endl;
cout << "Item 4 Price: $" << item4 << endl;
cout << "Item 5 Price: $" << item5 << endl;
cout << "Subtotal: $" << subtotal << endl;
cout << "Sales Tax: $" << salesTax << endl;
cout << "Total: $" << total << endl;
return 0;
}


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Output
Item 1 Price: $12.95
Item 2 Price: $24.95
Item 3 Price: $6.95
Item 4 Price: $14.95
Item 5 Price: $3.95
Subtotal: $63.75
Sales Tax: $3.825
Total: $67.575
item5;

Bonus Task:

Bonus Task

Break Number

```
#include <iostream>
using namespace std;
int main(){
 int num;
 cout <<" Enter a 5 digit integer: ";
 cin >> num;
 int d1 = num/10000;
 int d2 = (num/1000)% 10;
 int d3 = (num/100) % 10;
 int d4 = (num /10) % 10;
 int d5 = num % 10;
 cout << d1 <<"  "<< d2 <<"  "<< d3 <<"  "<<
  d4 <<"  " << d5 << endl;
 return 0;
 }
```

main.cpp

Run

```cpp
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a 5 digit integer:";
    cin >> num;
    int d1 = num / 10000;
    int d2 = (num / 1000) % 10;
    int d3 = (num / 100) % 10;
    int d4 = (num / 10) % 10;
    int d5 = num % 10;
    cout << d1 << "  " << d2 << "  " << d3 << "  " << d4 << "  " << d5 << endl;
    return 0;
}
```

Output

```
Enter a 5 digit integer:25000
2  5  0  0  0
```

Home Task

Task 1:

Home Task

Task 1

```
#include <iostream>
using namespace std;
int main(){
    int var1 = 10, var2 = 20, temp;
    cout << "Before swap: var1=" << var1 << ", var2=" 
    << var2 << endl;
    temp = var1;
    var1 = var2;
    var2 = temp;
    cout << "After swap: var1=" << var1 << ", var2=" 
    << var2 << endl;
    return 0;
}
```

C++ Online Compiler

main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    int var1 = 10, var2 = 20, temp;
    cout << "Before swap: var1 = " << var1 << ", var2 = "
<< var2 << endl;
    temp = var1;
    var1 = var2;
    var2 = temp;
    cout << "After swap:  var1 = " << var1 << ", var2 = "
<< var2 << endl;
    return 0;
}
```

Output

```
Before swap: var1 = 10, var2 = 20
After swap:  var1 = 20, var2 = 10
```

Task 2:

Task 2

```
#include <iostream>
using namespace std;
int main() {
    char lowerChar, upperChar;
    cout << "Enter a lowercase Alphabet (a-z): ";
    cin >> lowerChar;
    upperChar = lowerChar - 32;
    cout << "Uppercase alphabet: " << upperChar << endl;
    return 0;
}
```

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;
int main() {
    char lowerChar, upperChar;
    cout << "Enter a lowercase alphabet (a-z): ";
    cin >> lowerChar;
    upperChar = lowerChar - 32;
    cout << "Uppercase alphabet: " << upperChar << endl;
    return 0;
}

```

Crop

Output

```
Enter a lowercase alphabet (a-z): a
Uppercase alphabet: A
```